Exp No:14 Experiment to understand feature scaling.

Description: Understand the importance of feature scaling

Code:

import pandas as pd

df = pd.read_csv('/content/pre-process_datasample.csv')

df.head()

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes

df.Country.fillna(df.Country.mode()[0],inplace=True)

features=df.iloc[:,:-1].values

/tmp/ipykernel_1268/3424832005.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

df.Country.fillna(df.Country.mode()[0],inplace=True)

label=df.iloc[:,-1].values

from sklearn.impute import SimpleImputer

import numpy as np

age=SimpleImputer(strategy="mean",missing_values=np.nan)

Salary=SimpleImputer(strategy="mean",missing_values=np.nan)

age.fit(features[:,[1]])

Salary.fit(features[:,[2]])

SimpleImputer()

features[:,[1]]=age.transform(features[:,[1]])

features[:,[2]]=Salary.transform(features[:,[2]])

features

```
array([['France', 44.0, 72000.0],
       ['Spain', 27.0, 48000.0],
       ['Germany', 30.0, 54000.0],
       ['Spain', 38.0, 61000.0],
       ['Germany', 40.0, 63777.77777777778],
       ['France', 35.0, 58000.0],
       ['Spain', 38.77777777777778, 52000.0],
       ['France', 48.0, 79000.0],
       ['Germany', 50.0, 83000.0],
       ['France', 37.0, 67000.0]], dtype=object)
```

from sklearn.preprocessing import OneHotEncoder

oh = OneHotEncoder(sparse_output=False)

Country=oh.fit_transform(features[:,[0]])

Country

```
array([[1., 0., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [1., 0., 0.],
       [0., 0., 1.],
       [1., 0., 0.],
       [0., 1., 0.],
       [1., 0., 0.]])
```

final_set=np.concatenate((Country,features[:,[1,2]]),axis=1)

final_set

```
array([[1.0, 0.0, 0.0, 44.0, 72000.0],
       [0.0, 0.0, 1.0, 27.0, 48000.0],
       [0.0, 1.0, 0.0, 30.0, 54000.0],
       [0.0, 0.0, 1.0, 38.0, 61000.0],
```

[0.0, 1.0, 0.0, 40.0, 63777.77777777778],

[1.0, 0.0, 0.0, 35.0, 58000.0],

[0.0, 0.0, 1.0, 38.77777777777778, 52000.0],

[1.0, 0.0, 0.0, 48.0, 79000.0],

[0.0, 1.0, 0.0, 50.0, 83000.0],

[1.0, 0.0, 0.0, 37.0, 67000.0]], dtype=object)

from sklearn.preprocessing import StandardScaler

sc=StandardScaler()

sc.fit(final_set)

feat_standard_scaler=sc.transform(final_set)

feat_standard_scaler

array([[1.22474487e+00, -6.54653671e-01, -6.54653671e-01,

7.58874362e-01, 7.49473254e-01],

[-8.16496581e-01, -6.54653671e-01, 1.52752523e+00,

-1.71150388e+00, -1.43817841e+00],

[-8.16496581e-01, 1.52752523e+00, -6.54653671e-01,

-1.27555478e+00, -8.91265492e-01],

[-8.16496581e-01, -6.54653671e-01, 1.52752523e+00,

-1.13023841e-01, -2.53200424e-01],

[-8.16496581e-01, 1.52752523e+00, -6.54653671e-01,

1.77608893e-01, 6.63219199e-16],

[1.22474487e+00, -6.54653671e-01, -6.54653671e-01,

-5.48972942e-01, -5.26656882e-01],

[-8.16496581e-01, -6.54653671e-01, 1.52752523e+00,

0.00000000e+00, -1.07356980e+00],

[1.22474487e+00, -6.54653671e-01, -6.54653671e-01,

1.34013983e+00, 1.38753832e+00],

[-8.16496581e-01, 1.52752523e+00, -6.54653671e-01,

1.63077256e+00, 1.75214693e+00],

[1.22474487e+00, -6.54653671e-01, -6.54653671e-01,

-2.58340208e-01, 2.93712492e-01]])

from sklearn.preprocessing import MinMaxScaler

mms=MinMaxScaler(feature_range=(0,1))

mms.fit(final_set)

feat_minmax_scaler=mms.transform(final_set)

feat_minmax_scaler

```
array([[1.        , 0.        , 0.        , 0.73913043, 0.68571429],
       [0.        , 0.        , 1.        , 0.        , 0.        ],
       [0.        , 1.        , 0.        , 0.13043478, 0.17142857],
       [0.        , 0.        , 1.        , 0.47826087, 0.37142857],
       [0.        , 1.        , 0.        , 0.56521739, 0.45079365],
       [1.        , 0.        , 0.        , 0.34782609, 0.28571429],
       [0.        , 0.        , 1.        , 0.51207729, 0.11428571],
       [1.        , 0.        , 0.        , 0.91304348, 0.88571429],
       [0.        , 1.        , 0.        , 1.        , 1.        ],
       [1.        , 0.        , 0.        , 0.43478261, 0.54285714]])
```

Result:

Understood the importance of feature scaling